

Skill Link

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1 Introduction

India is home to a vast pool of skilled and semi-skilled manual workers across various sectors such as construction, carpentry, agriculture, plumbing, and electrical services. Despite their capabilities, many of these workers face significant barriers in accessing consistent and quality employment opportunities. These challenges arise due to a lack of digital literacy, limited internet penetration in rural areas, and the absence of a centralized platform that effectively connects them with potential employers [1][3].

On the other hand, businesses, households, and contractors often struggle to find trustworthy and available workers who match their specific requirements in terms of skills, location, and timing. This creates a demand-supply disconnect in the informal labor market, leading to inefficiencies and missed opportunities on both sides[2][4].

Skill Link is a proposed digital platform designed to bridge this critical gap. By leveraging modern web and mobile technologies, the platform aims to provide a user-friendly interface for skilled workers to register, showcase their profiles, and receive job alerts based on location and trade. For employers, it offers tools to post jobs, search for workers, and rate past services [2][3].

The project will also incorporate basic Artificial Intelligence (AI) and Machine Learning (ML) models to enhance user experience through smart recommendations, skill-tagging, and sentiment-based feedback systems. Additionally, the platform will support multilingual and voice-enabled features to ensure accessibility for semi-literate users in rural and semi-urban regions[4][5].

This initiative is not just a technological solution but a social impact project aimed at empowering the underserved workforce by bringing transparency, trust, and visibility to the informal job market[1][4][6].

2 Objectives

1. To learn and understand the existing challenges faced by skilled manual workers and job providers in the informal sector, including issues like digital illiteracy, lack of centralized job platforms, and communication barriers. design a responsive and user-friendly digital platform that allows seamless interaction between workers and employers through mobile and web applications, focusing on accessibility, simplicity, and multilingual support..
2. To develop Artificial Intelligence and Machine Learning-based modules that provide personalized job recommendations, sentiment-based feedback analysis, and trust-building mechanisms through verified profiles and ratings.
3. To analyze the system's performance, user feedback, and overall impact in improving job accessibility, transparency, and employment opportunities for skilled workers across various sectors.

3 Literature Review

Table 1: Summary of Key Literature Reviewed for the Skill Link Project

S.No	Source	Focus Area	Key Findings	Relevance to Project
1	PLFS Annual Report 2022-23 [1]	Employment statistics	Highlights lack of access to steady employment for skilled workers in rural/semi-urban areas	Justifies the need for a digital platform to increase job accessibility
2	AI-Driven Labor Platforms (Kaur & Singh, 2021) [2]	Use of AI in gig economy	AI-based recommendation systems improve job-worker matching efficiency	Supports integration of AI for smart job matching
3	NITI Aayog Report on Gig Economy (2023) [3]	Gig/platform economy trends	Identifies gaps in formal platforms for unorganized labor sectors	Backs the development of an inclusive labor marketplace like Skill Link
4	Digital Platforms for Informal Labor in India (Mahajan & Gupta, 2022) [4]	Role of tech in informal labor	Studies challenges in digital adoption for laborers and how tech platforms can bridge the gap	Validates Skill Link's approach to simplify onboarding and increase visibility for informal workers

The literature reviewed emphasizes the urgent need for platforms like *Skill Link*, combining the reach of mobile technology with the intelligence of AI to bridge employment gaps in India's skilled labor force. The key contributions of each source are elaborated below:

1. PLFS Annual Report 2022-23 [plfs2022]

This government report provides official labor statistics that highlight significant disparities in employment across rural and urban areas. It emphasizes that rural India has a surplus of skilled labor that remains underutilized. This supports the relevance of *Skill Link*'s mission to increase digital job access for these workers.

2. AI-Driven Labor Platforms (Kaur & Singh, 2021) [kaur2021aidriven]

This study explores the application of artificial intelligence in labor platforms, particularly in job recommendation systems and user feedback loops. It proves that AI can enhance user experience and trust by ensuring accurate job-worker matches—a central feature of *Skill Link*.

3. **NITI Aayog Report on Gig Economy (2023) [niti2023gigeconomy]**

This national-level report discusses the rapid rise of gig and platform-based employment in India. It identifies the structural absence of inclusive tools for informal sector workers and recommends scalable, mobile-first solutions. *Skill Link* addresses these gaps directly through its multilingual, location-aware, and AI-enabled platform.

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4 Methodology

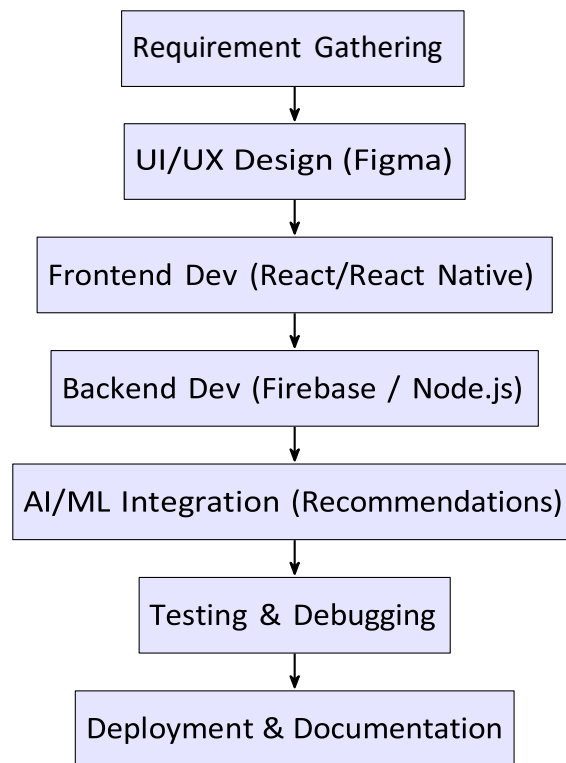


Figure 1: Project Development Flowchart

The methodology adopted for the development of Skill Link involves a structured and phased approach to ensure effective planning, implementation, and validation. The development process will be based on iterative prototyping and agile principles to accommodate feedback and evolving requirements. The platform will be developed as a web and mobile-based application, prioritizing performance, scalability, and user accessibility. Each phase in the development cycle contributes to ensuring that the platform is functionally robust, user-centric, and inclusive. The initial stages focus on requirement gathering through surveys, stakeholder interviews, and reviewing existing solutions. UI/UX design will emphasize simplicity, multilingual support, and voice-based navigation to cater to users with limited digital literacy.

During development, frontend technologies such as React.js or React Native will be used for seamless cross-platform deployment. Backend development will incorporate real-time databases and authentication services using Firebase or Node.js and MongoDB.

Machine learning components will be implemented using Python-based frameworks such as scikit-learn or TensorFlow Lite for mobile integration. These models will power

job matching, recommendation logic, and feedback analysis.

- UI design and mockups using Figma.
- Frontend development with React.js / React Native.
- Backend using Firebase / Node.js with MongoDB.
- Modules include registration, job posting, profile search, call/chat, and ratings.
- AI models for recommendations and sentiment-based feedback analysis.
- Voice interface integration for low-literacy accessibility.
- UI design and mockups using Figma.
- Frontend development with React.js / React Native.
- Backend using Firebase / Node.js with MongoDB.
- Modules include registration, job posting, profile search, call/chat, and ratings.

5 Project Plan and Timeline

Table 2: Detailed Timeline and Activity-wise Milestones for Skill Link

Activity / Months	July '25	August '25	September '25	October '25
Planning Phase				
Problem Identification	✓			
Requirement Analysis	✓			
Literature Review	✓			
Design Phase				
UI/UX Design (Figma)		✓		
System Architecture Final-ization		✓		
Development Phase				
Frontend Development (React)		✓	✓	
Backend Development (Firebase / Node.js)		✓	✓	
AI/ML Integration (Job Matching)			✓	
Testing and Evaluation				
Module Testing			✓	
User Feedback / Debugging			✓	✓
Reporting and Delivery				
Final Documentation				✓
Project Presentation				✓

6 Expected Outcomes

The successful implementation of the Skill Link platform is expected to generate a range of tangible and long-term benefits for both skilled manual workers and job providers. By addressing inefficiencies in the current labor market, the platform aims to make a meaningful social and economic impact, especially in underserved regions of India.

The anticipated outcomes include:

1. A fully functional, user-friendly digital platform (web and mobile) that connects skilled workers with job providers based on trade, location, and availability.
2. AI-driven personalized job recommendations and feedback analysis that improves the accuracy of job-worker matching over time.
3. Verified user profiles and a trust-building rating system to promote reliable, long-term employer-worker engagement.
4. Inclusion of accessibility features such as multilingual UI, voice-based interactions, and minimal text dependency to assist semi-literate users.
5. Enhanced employment opportunities and reduced job search time for skilled manual laborers in rural and semi-urban areas.
6. Potential scalability across regions and sectors, forming a blueprint for similar employment platforms targeting underserved worker populations.

7 Conclusion

The Skill Link project aims to bridge the gap between skilled manual workers and job providers through the use of smart digital technologies. By integrating Artificial Intelligence and Machine Learning, the platform enhances job matching accuracy and user trust.

This project not only provides technological innovation but also contributes to social empowerment by improving accessibility and transparency in the informal labor market. The inclusion of multilingual and voice-based features ensures that even semi-literate users can benefit from the platform.

In conclusion, Skill Link serves as a step toward building an inclusive and efficient employment ecosystem that supports both workers and employers, paving the way for sustainable digital transformation in the job market.